# Transformations

## STAT 223 – Applied Analytics

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# Transformations

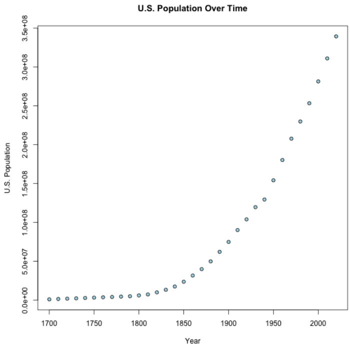
Linear regression relies on the data satisfying the assumptions, such as zero mean and homoscedasticity.

Obviously, in the real world, there are many cases where the data doesn't fit the data that well.

In these cases, transforming the data can result in a dataset that does satisfy the assumptions of linear regression.

# Transformations

For example, suppose I want to regress on the U.S. Population over time, shown on the graph below:




Clearly I can't do linear regression on this dataset.

# Transformations

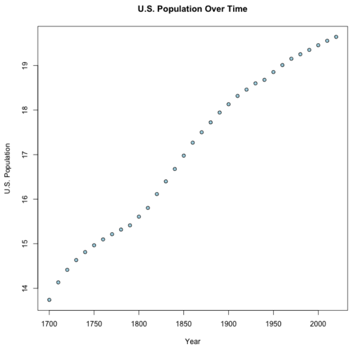
One of the simplest transformations we can do is applying the natural logarithm to our data.

$$Y = \beta_0 + \beta_1 \ln(X) + \varepsilon \qquad \text{OR} \qquad \ln(Y) = \beta_0 + \beta_1 X + \varepsilon$$

You can choose any base for the logarithm, but $e$ is the obvious choice.

# Transformations

Let's try applying the log transformation to the population data:




This is a lot more linear.

# Ladder of Powers

The **ladder of powers** is an orderly way of testing possible simple transformations. The transformation is expressed as:

$$Y = \beta_0 + \beta_1 X^\lambda + \varepsilon \qquad \text{OR} \qquad Y^\lambda = \beta_0 + \beta_1 X + \varepsilon$$

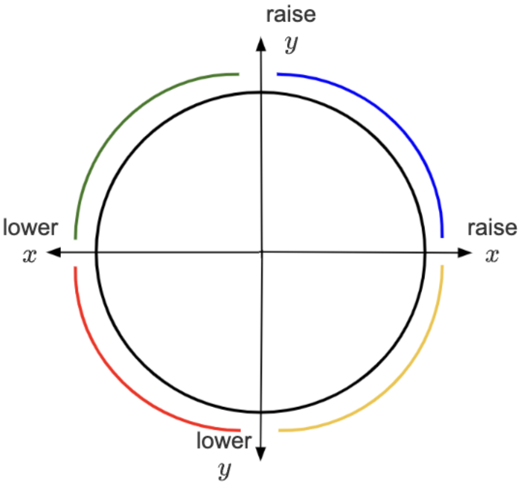
where $\lambda$ can take whatever value is necessary to make the relationship as close to linear as possible.

# Ladder of Powers

$$x(\lambda) = \begin{cases} x^{\lambda} & \lambda > 0 \\ \log(x) & \lambda = 0 \\ -x^{\lambda} & \lambda < 0 \end{cases}$$

# Ladder of Powers

$$Y = \beta_0 + \beta_1 X^{\lambda} + \varepsilon \qquad \text{OR} \qquad Y^{\lambda} = \beta_0 + \beta_1 X + \varepsilon$$



# Box-Cox Transformation

A similar procedure to the ladder of powers is the **Box-Cox transformation**. This is also commonly used.

$$x(\lambda) = \begin{cases} \dfrac{x^\lambda - 1}{\lambda} & \lambda \neq 0 \\ \ln(x) & \lambda = 0 \end{cases}$$

One nice thing is that you can use Python to find optimal values of $\lambda$ to get the best possible linear relationship.